

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1005

O'Fallon Renewable Energy Center 138 kV
July 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1005. The Interconnection Customer proposes a 959.1 MW natural gas combined cycle facility interconnecting at the O'Fallon Renewable Energy Center 138 kV substation in the Pennsylvania area. The request was received on May 30, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 255 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$78,254,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The CUEBRA 3801-O'FALLON RENEWABLE ENERGY CENTER 138 KV is overloaded to 437.2% of its 218 MVA emergency rating for the loss of the W HEMPFIELD-O'FALLON RENEWABLE ENERGY CENTER 138 KV. The project contributes 100.0% DFAX (959.1 MW) to this overload.

For the loss of O'FALLON RENEWABLE ENERGY CTR SUBSTATION-CULEBRA 3801 138 KV line, the O'FALLON RENEWABLE ENERGY CTR-WEST HEMPFIELD 138 KV loads to 427.1% of its 223 MVA rating. Project DFAX: 100.0%.

DNERVILLE-W HEMPFIELD 138 KV: 408.2% loading (230 MVA) following the outage of the CULEBRA 3801-O'FALLON RENEWABLE ENERGY CTR 138 KV L; DFAX = 100.0%.

The DONERVILLE SUBSTATION-ROCKY MOUNTAIN HYDROELECTRIC PLANT 138 KV is overloaded to 299.4% of its 204 MVA emergency rating for the loss of the CULEBRA 3801-O'FALLON RENEWABLE ENERGY CTR 138 KV. The project contributes 62.5% DFAX (599.5 MW) to this overload.

For the loss of CULEBRA 3801-O'FALLON RENEWABLE ENERGY CTR 138 KV, the RD-ROCKY MOUNTAIN HYDROELECTRIC PLANT 138 KV Line loads to 252.5% of its 230 MVA rating. Project DFAX: 62.5%.

MILLWOOD-RD 138 KV: 212.3% loading (204 MVA) following the outage of the CULEBRA 801-O'FALLON RENEWABLE ENERGY CENTER 138 KV line; DFAX = 41.6%.

The CRANE'S MILL-CULEBRA 3801 138 KV is overloaded to 187.1% of its 259 MVA emergency rating for the loss of the W HEMPFIELD-O'FALLON RENEWABLE ENERGY CENTER 138 KV. The project contributes 55.8% DFAX (535.1 MW) to this overload.

MILWOOD-ENGLESIDE SUBSTATION 138 KV: 173.2% loading (231 MVA) following the outage of the CULEBRA 3801-O'FALLON RENEWABLE ENERGY CTR 138 KV; DFAX = 41.6%.

IPL FIT MONROVIA I LLC-CRANE'S MILL 138 KV: 172.7% loading (275 MVA) following the outage of the WEST HEMPFIELD-O'FALLON RENEWABLE ENERGY CTR 138 KV; DFAX = 55.8%.

The RIVERSIDE MANUFACTURING-CULEBRA 3801 138 KV is overloaded to 170.1% of its 261 MVA emergency rating for the loss of the W HEMPFIELD-O'FALLON RENEWABLE ENERGY CENTER 138 KV. The project contributes 44.2% DFAX (424.0 MW) to this overload.

For the loss of O'FALLON RENEWABLE ENERGY CTR-WEST HEMPFIELD 138 KV, the ABBOT-RIVERSIDE MANUFACTURING 138 KV loads to 168.5% of its 246 MVA rating. Project DFAX: 44.2%.

IPL FIT MONROVIA I LLC-ENGLESIDE 138 KV: 151.3% loading (248 MVA) following the outage of the CULEBRA 3801-O'FALLON RENEWABLE ENERGY CTR 138 KV; DFAX = 41.6%.

For the loss of CULEBRA 3801-O'FALLON RENEWABLE ENERGY CENTER 138 KV, the GWF TRACY-DONERVILLE SUBSTATION 138 KV loads to 140.6% of its 228 MVA rating. Project DFAX: 37.5%.

For the loss of W HEMPFIELD SUBSTATION-O'FALLON RENEWABLE ENERGY CTR 138 KV, the ABBOT-POWELL 138 KV Line loads to 126.1% of its 267 MVA rating. Project DFAX: 20.0%.

GWF TRACY-ROHRERSTOWN 138 KV Line: 117.4% loading (247 MVA) following the outage of the CULEBRA 3801-O'FALLON RENEWABLE ENERGY CENTER 138 KV; DFAX = 37.5%.

The IPL FIT MONROVIA I LLC-KINGSTON 138 KV is overloaded to 104.4% of its 217 MVA emergency rating for the loss of the CONOWINGO 345-138 KV Auto. The project contributes 12.1% DFAX (116.2 MW) to this overload.

For the loss of ABBOT-POWELL 138 KV, the SHADOW GLEN-MRES ELBOW LAKE SUBSTATION 138 KV loads to 101.9% of its 172 MVA rating. Project DFAX: 11.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CUEBRA 3801-O'FALLON RENEWABLE ENERGY CENTER 138 KV	Rebuild line with higher-capacity conductor	\$4,143,000	100% (\$4,143,000)
O'FALLON RENEWABLE ENERGY CTR-WEST HEMPFIELD 138 KV	Rebuild line with higher-capacity conductor	\$4,893,000	100% (\$4,893,000)
DNERVILLE-W HEMPFIELD 138 KV	Rebuild line with higher-capacity conductor	\$2,431,000	100% (\$2,431,000)
DOERVILLE-ROCKY MTN HYDROELECTRIC PLANT 138 KV	Rebuild line with higher-capacity conductor	\$1,255,000	100% (\$1,255,000)
RD-ROCKY MOUNTAIN HYDROELECTRIC PLANT 138 KV Line	Rebuild line with higher-capacity conductor	\$2,196,000	100% (\$2,196,000)
MILLWOOD-RD 138 KV	Rebuild line with higher-capacity conductor	\$3,749,000	92% (\$3,429,000)
CRANE'S MILL-CULEBRA 3801 138 KV	Rebuild line with higher-capacity conductor	\$18,837,000	98% (\$18,506,000)
MLLWOOD SUBSTATION-ENGLESIDE 138 KV	Rebuild line with higher-capacity conductor	\$2,369,000	92% (\$2,167,000)
CRANE'S MILL-IPL FIT MONROVIA I LLC 138 KV	Rebuild line with higher-capacity conductor	\$4,393,000	98% (\$4,316,000)
RIVERSIDE MANUFACTURING-CULEBRA 3801 138 KV	Rebuild line with higher-capacity conductor	\$2,763,000	98% (\$2,702,000)
ABBOT-RIVERSIDE MANUFACTURING 138 KV	Rebuild line with higher-capacity conductor	\$5,871,000	98% (\$5,742,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
IPL FIT MONROVIA I LLC-ENGLESIDE 138 KV	Rebuild line with higher-capacity conductor	\$2,924,000	92% (\$2,675,000)
GWF TRACY-DONERVILLE 138 KV Line	Rebuild line with higher-capacity conductor	\$5,661,000	100% (\$5,661,000)
ABBOT SUBSTATION-POWELL 138 KV	Rebuild line with higher-capacity conductor	\$9,474,000	87% (\$8,204,000)
GWF TRACY-ROHRERSTOWN 138 KV Line	Rebuild line with higher-capacity conductor	\$2,054,000	100% (\$2,054,000)
IPL FIT MONROVIA I LLC-KINGSTON 138 KV	Reconductor line	\$6,250,000	82% (\$5,132,000)
SHADOW GLEN-MRES ELBOW LAKE SUBSTATION 138 KV	Reconductor line	\$3,321,000	83% (\$2,748,000)
	Total allocated to project		\$78,254,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.